

Meeting Summaries of Jan 15

Date: Jan 15, 2026

Time: 14:30–15:00 PM(PDT)

Attendees: Dr. Jaded, Dr. Yue and Feng

Topic: Feng's report on important parameters in partitioning environmental homogeneous groups; the underdeveloped final stage of piecewise quantile regression model

Summary of Discussion

- Feng delivered a brief discussion on the crucial parameters that influence the partition of environmental homogeneous groups under natural conditions and what the meanings of those parameters are.
- Feng reported the ongoing work in regression over **ZCI** and **Pollution Score (Sum-Rel)**, the two composite indices are derived from the PCs (Signals) decomposed from the matrices. These signal-wise correlations (species PCs vs pollution PCs) are expected to help construct a sensitively composite bioindicator (ZCI) that brings higher correlation with pollution levels (scores).
- Feng cross checked the results, like taxa composition and environmental characteristics of the bifurcations (which Feng had modified the cluster number (k) to 2 instead of 3 previously),
- Dr. Jaded wanted Feng to bring a **side-by-side comparison** between his results with the previous results from Zhang's work in 2008, expecting the exact or similar results of figures and tables. **On top of that**, Dr. Jaded expected Feng to **pinpoint** the differences or extensions of his work compared to Zhang's work.

Action Items

- Feng is going to set the proper parameters to reproduce Zhang's results on the hands on data, and bring the comparison to the next meeting.
- Feng is going to develop and embed the piecewise quantile regression model to complete the 'final' stage of the analysis pipeline.

Follow-Up

Temporary meeting time: January 22, 2026, 2:30-3:00 PM (PDT)

Supportive materials

- A meeting recording with subtitles has been restored on Google Drive for reviewing.

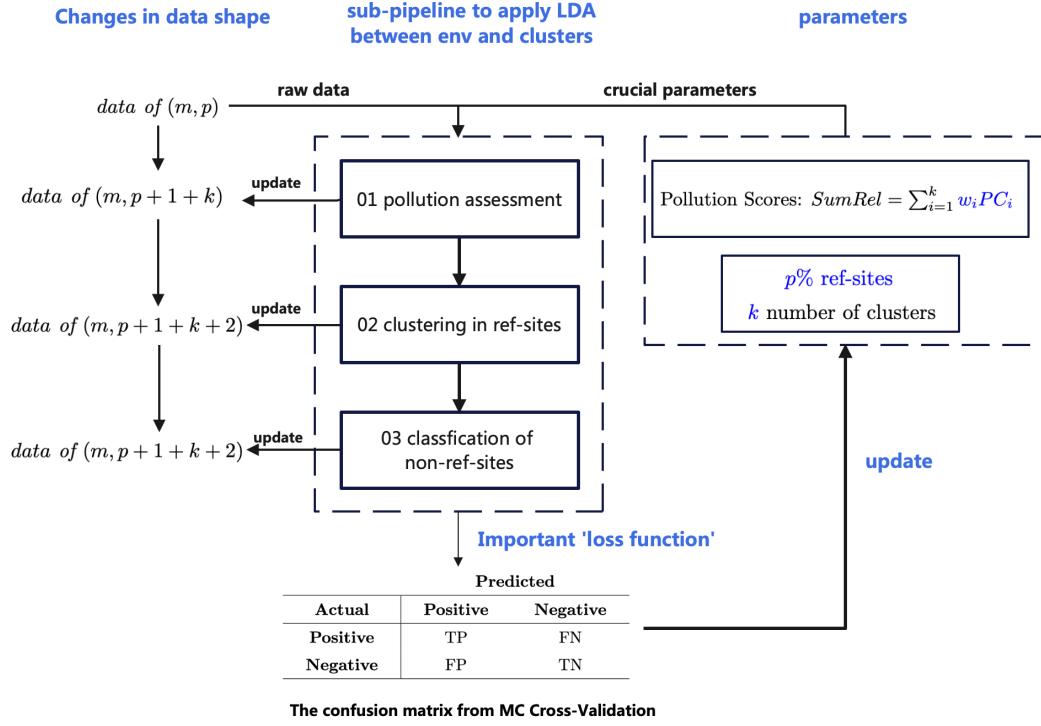


Figure 1: The important parameters to partition environmental homogeneous groups and the 'loss function' to optimize the partitioning.

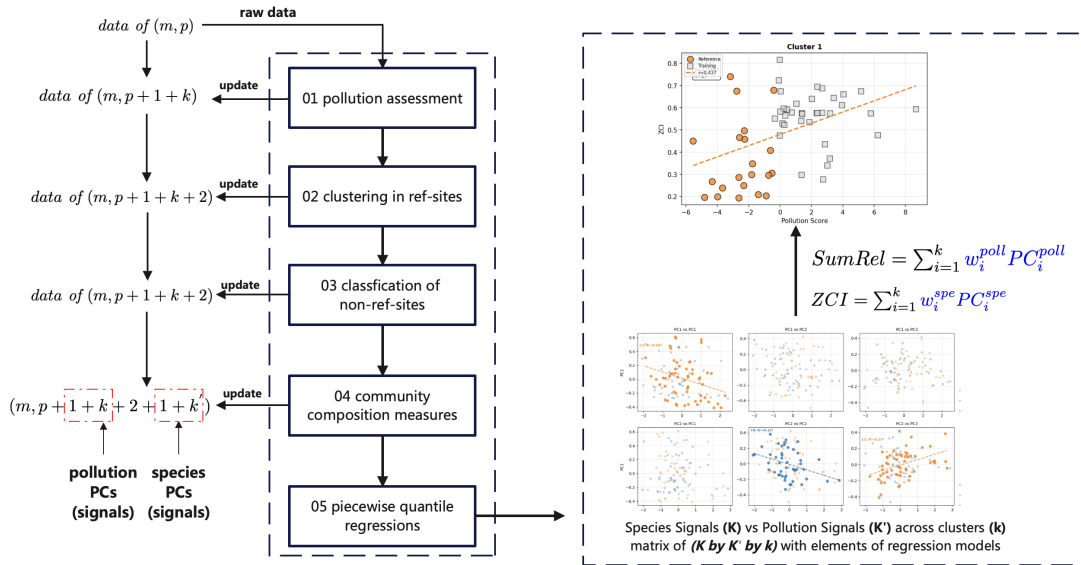


Figure 2: The built pipeline (up to stage 4) with the expected underdeveloped piecewise quantile regression model at the final stage (05).